0.1 Inner Product Space

Definition 0.1.1. Let V be a Vector Space over a field F, where $F = \mathbb{R}$ or $\mathbb{C}$. A map

$$\langle\cdot,\cdot\rangle : V \times V \to F : (x, y) \mapsto \langle x, y\rangle$$

is called *Inner product* of V if it satisfies the followings: for any $x, y, z \in V, \alpha \in F$,

1. $\langle \alpha \cdot x + y, z\rangle = \alpha \cdot \langle x, z\rangle + \langle y, z\rangle$. (sesquilinear)
2. $\overline{\langle x, y\rangle} = \langle y, x\rangle$. (conjugate)
3. $x \neq 0 \implies \langle x, x\rangle > 0$. (positiveness)

If a Space V has an inner product, then it is called *Inner product space*.

0.2 Normed Space

Definition 0.2.1. Let V be a Vector Space over a field F, where $F = \mathbb{R}$ or $\mathbb{C}$. A map

$$\|\cdot\| : V \to F : x \mapsto \|x\|$$

is called *Norm* of V if it satisfies the followings: for any $x, y \in V, \alpha \in F$,

1. $\|x + y\| \leq \|x\| + \|y\|$. (subadditivity)
2. $\|\alpha \cdot x\| = |\alpha| \cdot \|x\|$. (absolute homogeneity)
3. $\|x\| \geq 0$ and $\|x\| = 0 \iff x = 0$. (positive definiteness)

If a Space V has a Norm, then it is called *Normed space*.

Note: Inner Product can induce a Norm, and Norm can induce a Metric. That is,

$$\text{Inner Product Space} \subset \text{Normed Space} \subset \text{Metric Space}$$

Lemma 0.2.2. Let $\|\cdot\|$ be a Norm on a Vector Space V over $\mathbb{R}$ or $\mathbb{C}$. Then, for any $x, y \in V$,

$$|\|x\| - \|y\|| \leq \|x \pm y\| \leq \|x\| + \|y\|$$

Proposition 0.2.3. Let V be a Vector Space over $\mathbb{R}$ or $\mathbb{C}$.
If there exists an Inner product $\langle\cdot,\cdot\rangle$ on V, then a Norm induced by $\langle\cdot,\cdot\rangle$ as $\|\cdot\| \stackrel{\text{def}}{=} \sqrt{\langle\cdot,\cdot\rangle}$ satisfies:

$$\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2$$

(which is called *Parallelogram law*.) Conversely, fix a field as $\mathbb{R}$.
If there exists a Norm $\|\cdot\|$ on V which satisfies Parallelogram law, then the Norm can induce an Inner Product as:

$$\langle x,y\rangle \stackrel{\text{def}}{=} \frac{1}{4}\left[\|x+y\|^2 - \|x-y\|^2\right]$$

Proof. The first claim can be proved directly:

$$\begin{aligned}
\|x+y\|^2 + \|x-y\|^2 &= \langle x+y, x+y\rangle + \langle x-y, x-y\rangle \\
&= \langle x, x+y\rangle + \langle y, x+y\rangle + \langle x, x-y\rangle + \langle -y, x-y\rangle \\
&= \langle x,x\rangle + \langle x,y\rangle + \langle y,x\rangle + \langle y,y\rangle + \langle x,x\rangle + \langle x,-y\rangle + \langle -y,x\rangle + \langle -y,-y\rangle \\
&= 2\|x\|^2 + 2\|y\|^2
\end{aligned}$$

To prove the second claim, separates steps. We need to prove bilinear, conjugate, and positiveness.
Let $x,y,z \in V$ and $\alpha \in F$ be given.
Bilinear: the Parallelogram law gives that:

$$\begin{aligned}
\|x\|^2 + \|x-2y\|^2 = \|x-y+y\|^2 + \|x-y-y\|^2 = 2\|x-y\|^2 + 2\|y\|^2 \\
\|x+2y\|^2 + \|x\|^2 = \|x+y+y\|^2 + \|x+y-y\|^2 = 2\|x+y\|^2 + 2\|y\|^2
\end{aligned}$$

Substracting:

$$\|x+2y\|^2 - \|x-2y\|^2 = 2\|x+y\|^2 - 2\|x-y\|^2$$

Now,

$$\langle x, 2y\rangle = \frac{1}{4}(\|x+2y\|^2 - \|x-2y\|^2) = \frac{1}{2}(\|x+y\|^2 - \|x-y\|^2) = 2\langle x,y\rangle$$

Again, the Parallelogram law gives that:

$$\begin{aligned}
\|x+z+2y\|^2 + \|x-z\|^2 = \|x+y+(z+y)\|^2 + \|x+y-(z+y)\|^2 = 2\|x+y\|^2 + 2\|z+y\|^2 \\
\|x+z-2y\|^2 + \|x-z\|^2 = \|x-y+(z-y)\|^2 + \|x-y-(z-y)\|^2 = 2\|x-y\|^2 + 2\|z-y\|^2
\end{aligned}$$

Substracting:

$$\|x+z+2y\|^2 - \|x+z-2y\|^2 = 2(\|x+y\|^2 - \|x-y\|^2) - 2(\|z+y\|^2 - \|z-y\|^2)$$

Now,

$$\langle x+z, 2y\rangle = \frac{1}{4}\|x+z+2y\|^2 - \|x+z-2y\|^2 = \frac{1}{2}(\|x+y\|^2 - \|x-y\|^2) - \frac{1}{2}(\|z+y\|^2 - \|z-y\|^2) = 2\langle x,y\rangle + 2\langle z,y\rangle$$

From above discussion, for any natural number $n \in \mathbb{N}$,

$$\langle nx, y\rangle = \sum_{i=1}^{n} \langle x,y\rangle = n\langle x,y\rangle$$

This implies directly: for any $m \in \mathbb{N}$,

$$\langle x,y\rangle = \left\langle m\cdot\frac{1}{m}x, y\right\rangle = m\left\langle \frac{1}{m}x, y\right\rangle$$

Now, for any $p,q \in \mathbb{N}$ with $p \neq 0$,

$$\left\langle \frac{q}{p}\cdot x, y\right\rangle = \left\langle q\cdot\frac{1}{p}\cdot x, y\right\rangle = q\left\langle \frac{1}{p}\cdot x, y\right\rangle = \frac{q}{p}\left\langle x,y\right\rangle$$

It is easy to show that $\langle -x, y\rangle = -\langle x, y\rangle$, we have: for any rational $r \in \mathbb{Q}$,

$$\langle rx, y\rangle = r\langle x, y\rangle$$

Meanwhile, the subadditivity of the given norm gives:

$$\|x+y\|^2 \leq \|x\|^2 + \|y\|^2 + 2\|x\| \cdot \|y\|$$

and

$$\|x-y\|^2 \geq \|x\|^2 + \|y\|^2 - 2\|x\| \cdot \|y\|$$

Combining:

$$\|x+y\|^2 - \|x-y\|^2 \leq \|x\|^2 + \|y\|^2 + 2\|x\| \cdot \|y\| - (\|x\|^2 + \|y\|^2 - 2\|x\| \cdot \|y\|) = 4\|x\| \cdot \|y\|$$

Now,

$$\langle x, y\rangle = \frac{1}{4}\left[\|x+y\|^2 - \|x-y\|^2\right] \leq \|x\| \cdot \|y\|$$

The reverse inequality can be shown by similarly way. That is,

$$|\langle x, y\rangle| \leq \|x\| \cdot \|y\|$$

Let $r \in \mathbb{R}, q \in \mathbb{Q}$ be given. Then,

$$|r\langle x, y\rangle - \langle rx, y\rangle| \overset{(*)}{=} |(r-q)\langle x, y\rangle - \langle (r-q)x, y\rangle| \overset{(**)}{\leq} 2|r-q|\|x\| \cdot \|y\|$$

(The $(*)$ is held by the inner product preserves rational scaler product, and the $(**)$ is held by each term has same bound as $|r-q|\|x\| \cdot \|y\|$.)
Finally, let $\epsilon > 0$ be given. Then, for any $\in \mathbb{R}$, there exists $q \in \mathbb{Q}$ such that

$$|r\langle x, y\rangle - \langle rx, y\rangle| \leq 2|r-q|\|x\| \cdot \|y\| \leq \epsilon$$

by the Density of Rationals.
In conclusion, the inner product induced by a norm satisfying Parallelogram law satisfies bilinear.
Conjugate is clear, because we are in the Real numbers, and positiveness is clear too, by definition of Norm. $\square$

Note: Let V be a Vector Space over $\mathbb{C}$. Then, a restriction of the scalar multiplication $\cdot : V \times \mathbb{C} \to V$ to $\mathbb{R}$ still satisfies Vector Space Axiom. That is, V can be regarded as a vector space over $\mathbb{R}$.
The following theorems implicate that: For Complex Space V,

$V/\mathbb{C}$ has an Inner Product *if and only if* $V/\mathbb{R}$ has an Inner Product

Theorem 0.2.4. Suppose V is a Complex Inner Product Space with an Inner Product $\langle\cdot,\cdot\rangle : V \times V \to \mathbb{C}$. Consider V as a Vector Space over $\mathbb{R}$. Then, a map:

$$[\cdot,\cdot] : V \times V \to \mathbb{R} : (x,y) \mapsto \mathrm{Re}(\langle x,y\rangle)$$

is an Inner Product for $V/\mathbb{R}$ that satisfies for any $x \in V,\ [x, ix] = 0$.

Proof. Since inner product has three Axioms, we need three steps.
Sesqulinear: Let $\alpha \in \mathbb{R}$, $x, y, z \in V$ be given. Then,

$$[\alpha \cdot x + z, y] = \mathrm{Re}(\langle \alpha \cdot x + z, y\rangle) = \mathrm{Re}(\alpha \cdot \langle x,y\rangle + \langle z,y\rangle) = \alpha \cdot \mathrm{Re}(\langle x,y\rangle) + \mathrm{Re}(\langle z,y\rangle)$$

Conjugate and **Positiveness** are trivial. (Actually, there was one step!)
Meanwhile, for any $x \in V$, $\langle x,x\rangle$ is non-negative real number. That is,

$$[x, ix] = \mathrm{Re}(\langle x, ix\rangle) = \mathrm{Re}(i\langle x,x\rangle) = 0$$

□

Theorem 0.2.5. Suppose V is a Complex Inner Product Space. Let V be regarded as a Vector Space over $\mathbb{R}$. If $V/\mathbb{R}$ has an Inner Product $[\cdot,\cdot] : V \times V \to \mathbb{R}$ which satisfies for any $x \in V$, $[x, ix] = 0$, then, a map:

$$\langle\cdot,\cdot\rangle : V \times V \to \mathbb{C} : (x,y) \mapsto [x,y] + i[x,iy]$$

is an Inner Product for $V/\mathbb{C}$.

Proof. Again, we need three steps.
Sesqulinear: Let $a + ib \in \mathbb{C}$, $x, y, z \in V$ be given. Then,

$$\begin{aligned}
\langle (a+ib)\cdot x, y\rangle &= [(a+ib)\cdot x, y] + i \cdot [(a+ib)\cdot x, i\cdot y] \\
&= [a\cdot x, y] + [ib \cdot x, y] + i\cdot[a\cdot x, i\cdot y] + i\cdot[ib\cdot x, i\cdot y] \\
&= [a\cdot x, y] + i\cdot[a\cdot x, i\cdot y] + [ib\cdot x, y] + i\cdot[ib\cdot x, i\cdot y] \\
&= a\cdot[x,y] + i\cdot a\cdot[x, i\cdot y] + b\cdot[i\cdot x, y] + i\cdot b\cdot[i\cdot x, i\cdot y] \\
&= a\cdot\langle x,y\rangle + b\cdot[i\cdot x, y] + i\cdot b\cdot[i\cdot x, i\cdot y]
\end{aligned}$$

Meanwhile, since for any $x \in V$, $[x, ix] = 0$,

$$[x+y, i\cdot(x+y)] = [x, ix] + [y, ix] + [x, iy] + [y, iy] = 0 + [y, ix] + [x, iy] + 0 = 0$$

That is, for any $x, y \in V$, $[x, iy] = -[y, ix] = -[ix, y]$. This implies directly $[x,y] = [ix, iy]$. Now, succeed above process:

$$\begin{aligned}
b\cdot[i\cdot x, y] + i\cdot b\cdot[i\cdot x, i\cdot y] &= -b\cdot[x, i\cdot y] + i\cdot b\cdot[x,y] \\
&= i\cdot b\cdot([x,y] + i\cdot[x, i\cdot y]) \\
&= i\cdot b\cdot\langle x,y\rangle
\end{aligned}$$

In sum,

$$\langle (a+ib)\cdot x, y\rangle = a\cdot\langle x,y\rangle + b\cdot[i\cdot x, y] + i\cdot b\cdot[i\cdot x, i\cdot y] = a\cdot\langle x,y\rangle + ib\cdot\langle x,y\rangle$$

$\langle x+z, y\rangle = \langle x,y\rangle + \langle z,y\rangle$ is clear.
Conjugate:

$$\overline{\langle x,y\rangle} = [x,y] - i[x,iy] = [x,y] + i[ix,y] = [y,x] + i[y,ix] = \langle y,x\rangle$$

Positiveness is clear. □